# Amortized Bayesian parameter recovery for dynamical systems with conditional flow matching

The `amortix` package: architecture, evaluation methodology, and benchmarks

Ivan Oseledets

August 2026 (v0.1)

**Abstract**

We describe `amortix`, a package for amortized Bayesian estimation of the parameters of dynamical systems — ordinary and stochastic differential equations and discretized PDEs — from noisy observations at a finite set of time points. A conditional flow-matching model is trained once on simulated (parameter, data) pairs and thereafter produces posterior samples for any new observation set without likelihood evaluations or per-instance optimization; a single trained network handles any number and placement of observation points, the setting required for sequential experiment refinement and Bayesian experimental design. The package couples three components that we argue are only useful together: (i) a set-transformer encoder whose input representation is learnable rather than prescribed — a monotone mixture-of-scales reparametrization of the value axis, attention phases computed from physical observation times, and explicit set-cardinality features; (ii) a benchmark suite of fifteen systems spanning finance, neuroscience, epidemiology, ecology, pharmacokinetics, classical mechanics, and reaction–diffusion PDEs; and (iii) an evaluation harness in which simulation-based calibration is used only as a screen, and the instrument of record is comparison against exact or validated Markov chain Monte Carlo reference posteriors on the same observed points. On the fixed-design suite, 31 of 32 parameters across nine systems pass simulation-based calibration with a mean coverage error of 1.3 percentage points; on the variable-design suite, posterior widths match exact references to within 5–7% uniformly over design sizes $K = 5$–$100$, and the two residual defects found anywhere in the suite decay as measured power laws of the training budget. We compare against neural posterior estimation and flow-matching posterior estimation from the `sbi` package, both network families of BayesFlow 2, and a port of the Simformer, on three systems with exact or validated references. On a multiplicative process, all external configurations either inflate the volatility posterior by factors of 1.4–3.2 or become overconfident (widths 0.4–0.9) depending on the input coordinates they are given, while the learnable representation recovers the exact posterior from raw data. On an additive control process the same external tools improve markedly — the systematic bias disappears and the width inflation shrinks from 2.5–3.2 to 1.6–1.9 — which localizes the dominant failure to input conditioning while showing that a residual representation gap remains even in the benign case. We document this failure mechanism — the conditioning of the encoder input — in three distinct forms, together with learnable modules that repair each one.

## 1 Introduction

Recovering the parameters of a differential-equation model from noisy measurements of its solution is one of the basic inverse problems of applied science: a modeler writes down a dynamical system — an ODE, an SDE, or a PDE — whose right-hand side depends on unknown physical constants, observes a trajectory at a finite set of time points, and asks what the observations say about the constants. Like most inverse problems, this one is ill-posed in the practical sense: different parameter values can produce nearly indistinguishable observations, especially when the state is partially observed or the sampling is sparse, so a point estimate both hides the ambiguity and is unstable under noise. The Bayesian formulation resolves this by reporting the full posterior distribution over parameters given the data — ambiguity becomes posterior spread and correlation, quantities one can act on — at the price of two computational obstacles. First, for most stochastic and partially observed systems, the likelihood of the observations requires integrating over unobserved trajectory segments and is intractable or expensive. Second, modern workflows need posteriors *repeatedly*: calibration studies process thousands of synthetic observation sets, sequential experiments update the posterior after every new measurement, and Bayesian experimental design compares candidate observation schedules by the posteriors they would produce — so per-instance MCMC or optimization inside such loops is the dominant cost even when a likelihood is available.

Simulation-based inference (SBI) addresses both issues by learning a conditional generative model of the parameters given the data from simulated pairs — parameters drawn from the prior, observation sets produced by the simulator — thereby amortizing the cost of inference into a one-time training phase (Cranmer et al., 2020). Neural posterior estimation (NPE) and its sequential variants (Papamakarios and Murray, 2016; Greenberg et al., 2019) fit normalizing flows by maximum likelihood; flow-matching and diffusion variants (Wildberger et al., 2023; Gloeckler et al., 2024) replace the density estimator with transport-based generative models. Mature software exists: the `sbi` toolkit (Boelts et al., 2024), BayesFlow (Radev et al., 2020), and the `sbibm` benchmark collection (Lueckmann et al., 2021).

This report describes `amortix`, a compact PyTorch package (about 3,500 lines; one default model of $4.8 \times 10^5$ weights) for amortized Bayesian parameter recovery in dynamical systems. Three aspects of its design distinguish it from the toolkits above, and the report is organized around measuring their consequences.

The first concerns the interface between raw data and the network. Conditional density estimators are conventionally fitted to a fixed-length data vector after global standardization; the estimator sees whatever coordinates the user chose. We show in Sec. 8.2 that on multiplicative and heavy-tailed processes this convention fails in a specific, reproducible way across four different generative engines from three codebases, and that supplying hand-derived coordinates does not repair it uniformly: with identical inputs, different implementations end up over-dispersed, overconfident, or biased. In `amortix` the transformation from observations to network inputs is itself part of the trained model — a monotone mixture-of-scales reparametrization of the value axis, observation times entering attention as continuous phases, and explicit design-size features (Sec. 4); each component is motivated by a measured failure and validated by an ablation.

The second concerns observation designs. `amortix` trains a single network to approximate $p(m \mid S)$ coherently for every subset $S$ of the measurement points of an experiment — arbitrary sizes and placements, including spatio-temporal sensor designs for PDEs (Sec. 8.5). This is the object consumed by sequential refinement and Bayesian experimental design, and supporting it changes the training procedure (Sec. 5): designs must be resampled fresh at every optimizer step, and the distribution of design sizes becomes a modeling choice with measurable consequences.

The third concerns evaluation. Simulation-based calibration (Talts et al., 2018) is a necessary-condition test whose statistical power at practical sample sizes is limited; we measure that limit (Sec. 6) and use SBC only as a screen. The instrument of record throughout is the comparison of posterior mean and width against an exact or validated-MCMC reference posterior on the same observed points, and the package ships this machinery — closed forms, exact-likelihood adaptive Metropolis samplers, chain validation — as part of its public API. The concern motivating this discipline is documented in the SBI literature: learned posteriors can be confidently wrong in ways that rank diagnostics do not detect (Hermans et al., 2022).

The remainder of the report presents the method (Sec. 3), the architecture (Sec. 4), training procedures (Sec. 5), the evaluation methodology (Sec. 6), the benchmark suite (Sec. 7), experimental results including comparisons with `sbi`, BayesFlow, and the Simformer on three systems with exact or validated references (Sec. 8), an analysis of the failure modes encountered during development (Sec. 9), and limitations (Sec. 10). Appendix A defines every benchmark system with its equations, priors, and observation model; Appendix B details the reference posteriors; Appendix C maps the terminology of this report to package options.

# 2 Problem formulation

Let a dynamical system depend on a parameter vector $m \in \mathbb{R}^d$ with a known prior $p(m)$, uniform on a box in all benchmarks below. The system produces a state trajectory $x(t)$ — through an ODE $\dot{x} = f(x; m)$, an SDE $dx = f(x; m)\, dt + \sigma(x; m)\, dW_t$, or a spatially discretized PDE — and the trajectory is observed through a noisy channel at $K$ points:

$$y_i = h\big(x(t_i)\big) + \varepsilon_i, \qquad \mathcal{D} = \{(t_i, y_i)\}_{i=1}^{K}. \tag{1}$$

We call $\mathcal{D}$ the *observation set* of one experiment; it is the entire input to inference. Training and evaluation below involve many independently simulated observation sets, each paired with the parameter that generated it. The observation operator $h$ may expose only part of the state: in the benchmarks below, only the price of an asset while its stochastic volatility is latent, only the membrane voltage of a neuron while its gating variables are latent, or only three spatial sensors of a PDE field. The target is the posterior $p(m \mid \mathcal{D}) \propto p(\mathcal{D} \mid m)\, p(m)$.

Three standing assumptions delimit the setting. The parameter is finite- and low-dimensional ($d = 2$–5 in every benchmark below) with a prior that factorizes across components and has tractable marginal CDFs (uniform boxes throughout; the construction extends to any such prior). The simulator can be run forward cheaply and repeatedly, but nothing else is required of it — no likelihood evaluations, no gradients. Observation times (and sensor indices) are known exactly; the measurement noise $\varepsilon_i$ is independent across observations with a known per-system law.

**Running example.** The Ornstein–Uhlenbeck process is used throughout to make constructions concrete: $dX = -\theta X\, dt + \sigma\, dW$ with $m = (\theta, \sigma)$, prior box $[0.3, 3] \times [0.2, 1.5]$, stationary start $X_0 \sim \mathcal{N}(0, \sigma^2/2\theta)$. An observation set is, for instance, $K = 74$ noisy values of $X$ on a fixed two-rate grid, or — in the variable-design setting — any $K$ values at arbitrary times. Its posterior has the features that recur across the suite: the diffusion coefficient $\sigma$ is identified sharply by high-frequency increments; the reversion rate $\theta$ only weakly over a finite horizon; and the width of both marginals must shrink as observations accumulate. An estimator can therefore be wrong in the posterior mean, in the posterior width, or in the width-versus-$K$ trend, and the evaluation methodology of Sec. 6 is built to detect all three.

Although the posterior itself is intractable, sampling from the *joint* distribution of parameters and data is easy: draw $m_j \sim p(m)$, simulate the system, apply the observation channel. This yields arbitrarily many training pairs $(m_j, \mathcal{D}_j)$, and a conditional generative model trained on them approximates the posterior for every observation set simultaneously — the cost of inference is moved into a one-time training phase, and inference for a new observation set is a forward pass. We impose one requirement beyond standard amortization: a single trained network must remain correct for any number $K$ and any placement of observation times (and, for PDEs, sensor positions). Formally, for every subset $S$ of the measurement points of one experiment the network must approximate $p(m \mid S)$, and the family of answers must be coherent — in particular the posterior must widen as points are removed. We refer to this requirement as *design amortization*; it is the object consumed by sequential refinement and by Bayesian experimental design, where candidate observation schedules are ranked by the posteriors they would produce.

# 3 Posterior sampling by conditional flow matching

The idea of the method is to represent the posterior not by a density formula but by a *transport*: a learned, data-conditioned deformation that carries a fixed reference distribution onto the posterior. Sampling then amounts to drawing from the reference and applying the deformation — there is no normalizing constant to compute and no invertibility constraint to build into the network, because the deformation is defined implicitly as the flow of a velocity field, and the field itself is fitted by ordinary least-squares regression on simulated pairs. This construction is conditional flow matching (Lipman et al., 2023).

Concretely, parameters are first reparametrized componentwise,

$$m \;\longmapsto\; \Phi^{-1}\big(F_{\text{prior}}(m)\big) \in \mathbb{R}^d, \tag{2}$$

where $F_{\text{prior}}$ collects the marginal prior CDFs and $\Phi$ is the standard normal CDF. In the new coordinates the prior is exactly $\mathcal{N}(0, I)$ and box constraints disappear; usefully, the uninformative limit then costs nothing — when the data carry no information the posterior coincides with the reference and the transport is the identity. From here on we work exclusively in these coordinates and keep writing $m$ for the parameter in them; the inverse of (2) is applied exactly once, at the end of sampling, to return draws on the original scale.

The transport is indexed by a *virtual time* $\tau \in [0, 1]$ — an algorithmic variable with no relation to the physical time $t$ at which the dynamical system is observed; $t$ appears only inside observation sets, $\tau$ only in the transport. Subscripts in $\tau$ denote the position along the transport path: $m_1$ is a parameter drawn jointly with its observation set $\mathcal{D}$ (the endpoint), $m_0 \sim \mathcal{N}(0, I)$ is pure reference noise — not the transform of any parameter, but the raw material that the transport will have turned *into* a posterior draw by $\tau = 1$ — and between them lies the training interpolation $m_\tau = (1 - \tau)\, m_0 + \tau\, m_1$. The velocity network is trained to predict the direction of this path by regression:

$$\mathcal{L}(\theta) = \mathbb{E}_{m, \mathcal{D},\; m_0 \sim \mathcal{N}(0,I),\; \tau \sim U[0,1]} \big\| v_\theta(m_\tau,\, \tau,\, \mathcal{D}) - (m_1 - m_0) \big\|^2. \tag{3}$$

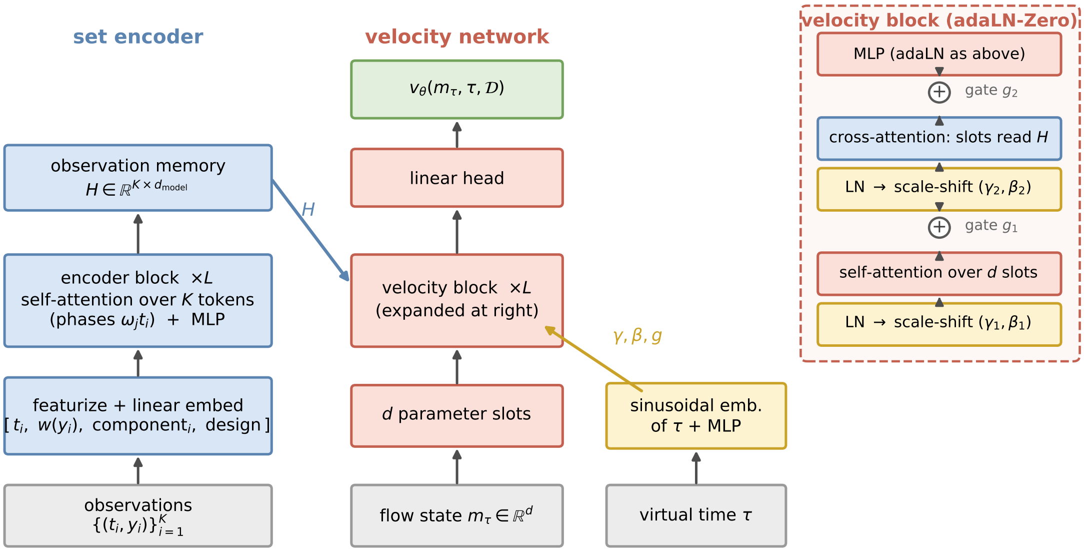


**Figure 1:** Model architecture; arrows denote data flow, bottom to top. Left: each observation is featurized and linearly embedded into one token; $L$ transformer encoder blocks, whose attention phases are computed from the physical observation times (Sec. 4.2), produce a memory $H$ of $K$ encoded states. Middle: the $d$ components of the flow state $m_\tau$ form $d$ parameter-slot tokens processed by $L$ velocity blocks and a linear head into the velocity $v_\theta$. Right: one velocity block expanded — self-attention across parameter slots, cross-attention reading the observation memory $H$, and an MLP, each preceded by layer normalization with scale-shift $(\gamma, \beta)$ and followed by a gated residual connection ($g$, marked +); the triples $(\gamma, \beta, g)$ are produced from the virtual time $\tau$ by a zero-initialized map, so the untrained block is the identity (Peebles and Xie, 2023).

The minimizer of (3) is the marginal velocity field whose flow transports $\mathcal{N}(0, I)$ at $\tau = 0$ to $p(m_1 \mid \mathcal{D})$ at $\tau = 1$ (Lipman et al., 2023). Sampling draws $m_0 \sim \mathcal{N}(0, I)$, integrates $dm/d\tau = v_\theta(m, \tau, \mathcal{D})$ from $\tau = 0$ to 1, and maps the endpoint back through the inverse of (2); the numerical settings of the integration are implementation details, collected in Sec. 5. The same estimator family, with a different conditioning architecture, is available in `sbi` as FMPE (Wildberger et al., 2023); the comparison of Sec. 8.2 therefore isolates the effect of the input representation from the choice of generative engine.

No likelihood evaluations, simulator gradients, or per-instance optimization are required at any stage; the simulator is executed only forward, during training-set generation.

The training objective is consistent in the following sense.

**Proposition 1** (consistency)**.** *Let $(m, \mathcal{D}) \sim p(m)\, p(\mathcal{D} \mid m)$ and let $m_1$ be given by (2). If the loss (3) is minimized over all measurable velocity fields, then for $p(\mathcal{D})$-almost every $\mathcal{D}$ the flow of the minimizer transports $\mathcal{N}(0, I)$ at $t = 0$ to the conditional law $p(m_1 \mid \mathcal{D})$ at $t = 1$; equivalently, the sampling procedure above draws exactly from the posterior (Lipman et al., 2023; Wildberger et al., 2023).*

The proposition is an idealization: with a finite network, a finite simulation budget, and a numerical solver, no calibration guarantee survives, and the residual error becomes an empirical quantity. Section 6 builds the instruments that measure it.

# 4 Conditioning architecture

The velocity field $v_\theta(m, \tau, \mathcal{D})$ must be conditioned on an observation set of time-stamped measurements, and three properties of that set drive the architecture: the number of observations $K$ varies from experiment to experiment (by two orders of magnitude in the variable-design setting); the observation times are arbitrary real numbers, in general not equally spaced, so positional schemes built for uniform sequences do not apply; and the times carry all ordering information, so the output must not depend on how the observations happen to be stored. These properties rule out the fixed-length-vector conditioners that are the default in NPE implementations, and they are naturally met by a transformer operating on one feature vector per observation — throughout this section, a *token*.

## 4.1 Overall scheme

The network evaluates $v_\theta(m, \tau, \mathcal{D})$, where $m$ is the normalized parameter vector partway along its transport path — the interpolation $m_\tau$ at training time, the current ODE state at sampling time — and $\tau$ is the virtual time of the transport. The design follows the encoder–decoder pattern that is standard in conditional flow-matching and diffusion models: the data are encoded once, and a decoder combines that encoding with the object being generated,

$$v_\theta(m, \tau, \mathcal{D}) \;=\; \mathrm{Dec}\big(m,\ \tau,\ \mathrm{Enc}(\mathcal{D})\big), \tag{4}$$

with explicit types: Enc maps the observation set — the $K$ time-stamped measurements of one experiment — to a memory of $K$ vectors, $\mathrm{Enc}(\mathcal{D}) \in \mathbb{R}^{K\times d_{\mathrm{model}}}$, computed once per observation set, and $\mathrm{Dec} : \mathbb{R}^d \times [0,1] \times \mathbb{R}^{K\times d_{\mathrm{model}}} \to \mathbb{R}^d$ combines the current transport point and the virtual time with that memory. Two differences from the image and text instances of this pattern dictate the concrete form (Fig. 1). First, the conditioning is not a fixed-size embedding but an observation set whose size $K$ varies: Enc is therefore a transformer over per-observation tokens, its output is a *memory* of $K$ encoded states rather than one pooled vector, and Dec reads the memory by cross-attention — the one standard mechanism indifferent to $K$. Second, the generated object is not an image but a short vector of $d$ parameters, so the decoder is small: the $d$ components of $m$ are treated as $d$ tokens, self-attention among them expresses posterior correlations (the ridge of the Fisher–KPP posterior, the drift–volatility coupling of GBM), and the virtual time $\tau$ enters every block through zero-initialized adaptive layer normalization (adaLN-Zero), the conditioning mechanism of diffusion transformers (Peebles and Xie, 2023), so that the untrained network is an identity-like map.

On the encoder side, observation $i$ is described by a few features — its time $t_i$, its transformed value $w(y_i)$ with $w$ the learnable monotone map of Sec. 4.3, an index identifying the sensor or state component that produced the value (multivariate states contribute one token per observed component per time), and the design-size feature of Sec. 4.3 — and lifted to the model width by a learned linear map. The attention mechanism that lets these tokens interact is the subject of the next subsection; the choice of features is the subject of Sec. 4.3.

This scheme was not the first attempt but the survivor of a sequence of measured alternatives; the failed ones are retained in the package as reproducible controls and are compared under a common protocol in the ablation study of Sec. 8.5. All sizes are configuration parameters. One default configuration (width 64, four heads, three blocks per side) is used unchanged for every experiment in this paper; at the training-set sizes we use, models of this scale proved sufficient — doubling width and depth changes no benchmark result (Sec. 8.4) — and nothing in the construction caps the scale.

## 4.2 Attention phases from observation times

We adapt rotary position embeddings — the standard positional scheme of modern transformers (Su et al., 2024) — to irregular time sampling.

First the classical construction. Let $x_1, \ldots, x_K \in \mathbb{R}^{d_{\mathrm{model}}}$ be the token states entering an attention layer. Each head computes queries, keys, and values by learned linear maps,

$$q_i = W_Q x_i, \qquad k_i = W_K x_i, \qquad u_i = W_V x_i, \qquad W_Q, W_K, W_V \in \mathbb{R}^{d_h \times d_{\mathrm{model}}}, \tag{5}$$

and token $i$ aggregates the others with weights $a_{il} = \mathrm{softmax}_l\big(\hat{q}_i^\top \hat{k}_l/\sqrt{d_h}\big)$ applied to the $u_l$, where $\hat{q}, \hat{k}$ are position-modulated versions of $q, k$ defined as follows. Split each query and key into $d_h/2$ two-dimensional blocks, $q_i = (q_i^{(1)}, \ldots, q_i^{(d_h/2)})$ with $q_i^{(j)} \in \mathbb{R}^2$, and rotate block $j$ by an angle $\phi_i$ proportional to a per-token scalar:

$$\hat{q}_i^{(j)} = R(\omega_j \phi_i)\, q_i^{(j)}, \qquad \hat{k}_i^{(j)} = R(\omega_j \phi_i)\, k_i^{(j)}, \qquad R(\phi) = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}. \tag{6}$$

In the classical scheme the scalar is the token's integer position in the sequence, $\phi_i = n_i$, and the frequencies form a geometric ladder across blocks, $\omega_j = \omega_{\max}(\omega_{\min}/\omega_{\max})^{(j-1)/(d_h/2-1)}$, so that high-frequency blocks resolve neighboring positions and low-frequency blocks long ranges. Because rotations preserve inner products up to the relative angle,

$$\hat{q}_i^\top \hat{k}_l \;=\; \sum_{j=1}^{d_h/2} \big(q_i^{(j)}\big)^\top R\big(\omega_j(\phi_l - \phi_i)\big)\, k_l^{(j)}, \tag{7}$$

every attention logit depends on the positions only through differences $\phi_l - \phi_i$ — relative position awareness with no positional parameters.

For irregularly sampled data the integer position is meaningless, and the generalization is immediate: we set $\phi_i = t_i$, the physical time stamp of observation $i$, and choose the frequency range so that the periods $2\pi/\omega_j$ span from the finest observable time gap to several times the horizon. Nothing else changes; by (7), every logit now depends on the time stamps only through differences $t_l - t_i$. The substitution of physical time for position is the same one that time-aware attention models for irregular clinical and event series make in other positional schemes (Horn et al., 2020; Shukla and Marlin, 2021). Two consequences follow at once: the encoder is exactly permutation-invariant (relabeling tokens leaves the right-hand side of (7) unchanged; verified numerically to $10^{-8}$), and variable designs require no positional bookkeeping. The phases enter only the encoder's self-attention; the cross-attention from parameter slots to the observation memory is unmodified attention, since parameter slots carry no time stamp and the memory states already encode temporal structure.

The adaptation is also motivated by a measured failure. An earlier configuration used integer-position rotations, trained with long token sequences subsampled in place (kept tokens retained their spread-out positions), and evaluated on contiguously packed short sequences. Every component was individually correct, yet the mismatch between the positional distributions seen at training and evaluation produced systematic posterior biases of +1.1 to +3.3 standard deviations (measured in Sec. 8.5). Integer positions create an implicit train/test contract on token placement; physical-time phases eliminate the contract rather than manage it.

## 4.3 Input features

The remaining design choices concern what the tokens contain. Three of them matter and are described in turn: a learnable transform of the observed values, an optional increment-based token construction for directly observed diffusions, and an explicit design-size feature. Each occupies a few lines of code; each is load-bearing, as the ablations of Sec. 8.5 show.

**Value transform.** State variables of the benchmark systems live on very different scales, and some — prices, concentrations, populations — are multiplicative: the information sits in relative rather than absolute changes, and the coordinate in which such a variable is well-behaved is logarithmic. Rather than prescribing coordinates per problem, each value enters its token through a learned monotone map,

$$w(x) \;=\; \sum_{k=1}^{S} a_k \,\operatorname{sign}(x) \log\!\big(1 + |x|/s_k\big), \qquad a_k \geq 0. \tag{8}$$

Each basis function is linear for $|x| \ll s_k$ and logarithmic for $|x| \gg s_k$, so the family contains the identity, the signed logarithm, and every compression in between; positivity of the weights makes $w$ monotone by construction; and — the operative property — every candidate compression is present from initialization, so training only reweights fixed nonlinearities, and the downstream network can exploit whichever coordinate helps from the first optimizer step. The scales $s_k$ sit on a fixed geometric grid covering the dynamic range of the data ($S = 8$ octaves by default, one setting for all fifteen systems, no per-problem adjustment).

This structure has to earn its place against simpler-looking learnable alternatives, and we tested them on GBM, where the correct coordinate (log-price) is known and the cost of missing it is a measured +0.48 sd volatility bias. A generic two-layer MLP on the raw values, and a random-Fourier-feature map, both left the bias in place at matched budget: a free-form map must discover the shape and the scale of the compression simultaneously, from gradients arriving through the whole downstream network, and in practice it does not. A classical one-parameter statistical family — the Yeo–Johnson power transforms (Yeo and Johnson, 2000), in essence $x \mapsto ((1 + |x|)^\lambda - 1)/\lambda$ applied with sign symmetry, which passes through the identity at $\lambda = 1$ and tends to the logarithm as $\lambda \to 0$ — also failed, in an instructive way: trained end-to-end, $\lambda$ never left its initialization, because a single global exponent must traverse a gradient plateau before the downstream network has learned anything that would reward the move. The mixture (8) removes the bias reliably, and on GBM the trained weights concentrate on the small-scale components: the model learns the log-price coordinate.

**Increment features for directly observed diffusions.** For scalar SDEs whose state is observed directly — no latent components — the package offers a second token construction: after sorting by time, consecutive observations contribute

$$\big[\, t_{(i)},\ w(y_{(i)}),\ \Delta w_i,\ (\Delta w_i)^2,\ w_t(\Delta t_{(i)}) \,\big], \qquad \Delta w_i = w(y_{(i+1)}) - w(y_{(i)}).$$

The motivation is elementary: for a Markov observed process the likelihood factorizes over consecutive pairs, transition densities of diffusions depend on increments and gaps, and squared increments carry the quadratic-variation information that identifies diffusion coefficients. This is an inductive bias of the same character as convolutions for images — useful where its assumption holds, not a requirement of the method: the point-token path of Sec. 4.1 is fully general, and the ablations of Sec. 8.5 quantify what the increment tokens buy at fixed budget.

The rule for choosing between the constructions — increment tokens exactly when the observed process is Markov — was tested in both directions: an increment-token model class transfers across Markov systems without adjustment (GBM to OU), and on the Heston model, whose observed price is not Markov because volatility is latent, increment and point tokens perform identically across all 25 evaluation cases, as the factorization argument predicts. The package selects the construction from a per-problem flag.

Two further normalization details matter for processes with jumps, where squared increments within a single observation set span several orders of magnitude and a global feature normalization becomes unstable — concretely, on the Merton benchmark the running normalization scale is set by rare jump outliers and the resulting representation drift produces a +1.12 sd volatility bias. First, increment features are logarithmically compressed before normalization; this is a near-identity on systems without jumps (since $\log(1 + x) \approx x$ for small $x$) and by itself turns the Merton failure into a calibrated posterior across all 25 evaluation cases (Sec. 9). Second, the normalization is made adaptive to the observation set: robust summary statistics $s$ of the observation set, pooled over non-padding positions, are mapped by a zero-initialized MLP to per-channel scale and shift corrections,

$$h_c \ \mapsto\ \big(1 + \gamma_c(s)\big)\, h_c + \beta_c(s) \tag{9}$$

— a feature-wise affine modulation, FiLM in the terminology of Perez et al. (2018). Its measured contribution is specific and modest: it removes a residual +0.27 sd bias that survives on the densest Merton designs (the effective jump-detection threshold shifts with the unknown volatility), and it does so as well as a hand-crafted per-observation-set rescaling, within error bars ($+0.13 \pm 0.06$ vs $+0.09 \pm 0.05$ sd; Sec. 9). Three implementation conditions are necessary and were each found through a failure: zero initialization (training starts from the working baseline), modulation applied *after* the running normalization (the reverse ordering re-creates the drift it corrects), and pooled statistics that exclude padding positions.

**Design size.** Posterior width is governed by the number of observations — in regular models the error scales as $K^{-1/2}$ — but $K$ is a global property of the set, and softmax attention, which computes normalized weighted means, is nearly invariant to it. Softmax attention computes normalized weighted means, which are nearly invariant to the number of tokens — yet posterior width is governed by $K$. The deficit is measurable by linear probes from the pooled encoder output: all architecture variants of Table 7 recover posterior-mean statistics ($R^2 = 0.83$–$0.94$), while $R^2(\log K)$ ranges from 0.985 to 0.498 and ranks the variants exactly in the order of their width errors. The repair is to make the count available, and the package default is the simplest form: $\log K$ enters every token as one of the input features. An alternative that requires no explicit feature — appending a few learned auxiliary tokens to the set, in the spirit of register tokens (Darcet et al., 2024), and letting the model accumulate the count in them — achieves the same effect in our ablations; the number of such tokens is a hyperparameter with no measurable influence beyond one. With the count available, the width error on the hardest case ($K = 100$) drops from 1.95 to 1.49 with bias unchanged; the residual is closed by the training procedure of the next section.

# 5 Training procedures

**Common settings.** Sampling integrates the velocity field with a fixed-step midpoint scheme, 20 steps by default; the step count was fixed once by a convergence check (fourth-order Runge–Kutta with three times as many steps changes no benchmark statistic), and evaluation uses an exponential moving average of the training weights. All models are trained with Adam at a base learning rate of $3 \times 10^{-4}$ under a cosine schedule with linear warmup, batch size 64, and gradient clipping; evaluation uses an exponential moving average of the weights with decay 0.999. At every optimizer step the virtual times $\tau$ are drawn afresh (stratified over $[0, 1]$) together with fresh reference noise $m_0$, so no pairing between observation sets and virtual times is ever repeated. For fixed-design problems the training set consists of $n$ simulated observation sets and training runs for a fixed number of epochs over it; the gallery of Sec. 8.1 uses $n = 40{,}000$ simulations and 35 epochs.

**Design amortization.** Training data are simulated once on a fine grid; observation designs are then resampled *from the stored fine trajectories at every optimizer step.* The procedure has three components, each fixed by a controlled comparison on three systems (GBM, Ornstein–Uhlenbeck, FitzHugh–Nagumo):

1. *Fresh designs at every step.* Re-drawing each trajectory's observation subset at each step makes design memorization impossible and lets one simulation serve arbitrarily many designs. The alternative — one fixed design per trajectory — closes the dense-design error at long budgets but overfits the design set, producing overconfidence at sparse $K$ (width ratio 0.86 against the exact reference).
2. *The design-size law.* $K$ is drawn 50% log-uniform on $[K_{\min}, K_{\max}]$ and 50% uniform on the dense half of the range. Log-uniform sampling alone places $\approx 6\%$ of the mass at $K \geq 100$ and starves the dense regime, which appears as inflated widths precisely there; the mixture alone halves the dense-design excess (GBM $1.27 \to 1.14$, OU $1.34 \to 1.14$; the FitzHugh–Nagumo dense-$K$ SBC failure at $p < 10^{-3}$ disappears, $p = 0.54$).
3. *Budget scaling.* When higher accuracy is required, simulations and optimizer steps are scaled together at constant epochs; Sec. 8.4 shows the residuals then follow power laws in this single budget variable.

With this procedure, the GBM posterior width matches the exact per-design posterior to 5–7% uniformly over $K = 5, 9, 20, 100$ (Fig. 5).

# 6 Evaluation methodology

Two instruments with different roles are used throughout, plus a diagnostic probe.

## 6.1 Simulation-based calibration as a screen

SBC (Talts et al., 2018) tests a necessary condition: for $m^* \sim p(m)$ and $\mathcal{D} \sim p(\mathcal{D} \mid m^*)$, the rank of each component of $m^*$ among posterior samples must be uniform. We run SBC per parameter (uniformity test on 500 simulated observation sets with 200 posterior draws each unless stated; $p > 0.05$ passes) and report coverage of central 50% and 90% intervals.

The power of the test at this size is limited, and because our conclusions depend on knowing that limit, we measured it: on cases where an exact reference is available, SBC at 300–500 observation sets fails to reject posterior width ratios up to $\approx 1.3$, and its per-parameter $p$-values are unstable between retrainings when residual biases are in the range 0.1–0.2 posterior standard deviations. SBC also cannot penalize a posterior that ignores the data and returns the prior; the prior-mean and ridge controls of Sec. 7 exist for that reason. SBC is therefore used as a cheap screen, never as a verdict.

## 6.2 Reference posteriors as the record

Wherever a tractable likelihood exists, we compare distributions on the same observed points using two statistics per parameter: the signed error of the posterior mean in units of the reference posterior standard deviation (*bias*) and the ratio of standard deviations (*width*; 1 is exact, $> 1$ conservative, $< 1$ overconfident). References are of three kinds (full details in Appendix B):

- *Closed form.* The linear-Gaussian benchmark (exact Gaussian posterior) and geometric Brownian motion, whose log-returns give a conjugate normal–inverse-$\chi^2$ posterior importance-corrected to the box prior.
- *Exact-likelihood MCMC.* An adaptive random-walk Metropolis sampler (after Haario et al., 2001): proposal scale adapted to a 0.25 acceptance target during burn-in, proposal shape replaced by the empirical covariance Cholesky factor rescaled to $2.38/\sqrt{d}$, adaptation frozen before the retained phase, out-of-box proposals rejected (which samples the box-truncated target exactly). Likelihoods are exact per-gap transition densities: Gaussian Euler–Maruyama chain transitions for OU (matching the simulator that generated the data, plus the stationary density of the informative initial

| system | type | $d$ | obs. | design | reference posterior |
|---|---|---|---|---|---|
| linear-Gaussian | linear map | 4 | full | fixed | exact Gaussian |
| Ornstein–Uhlenbeck | SDE | 2 | full | fixed | exact-likelihood MCMC |
| geometric Brownian motion | SDE | 2 | full | fixed | conjugate closed form |
| Cox–Ingersoll–Ross | SDE | 3 | full | fixed | — (Euler pseudo-MLE) |
| double well | SDE | 3 | full | fixed | — |
| stochastic Lotka–Volterra | 2-d SDE | 4 | full | fixed | — |
| FitzHugh–Nagumo | 2-d SDE | 4 | $v$ only | fixed | — |
| SEIR | 4-d SDE | 5 | $I, D$ | fixed | — |
| polynomial-drift SDE | SDE | 5 | full | fixed | — |
| Heston | 2-d SDE | 5 | price only | variable | — |
| Merton jump-diffusion | jump SDE | 5 | full | variable | Poisson-mixture MCMC |
| Hénon–Heiles | Hamiltonian ODE | 3 | $q_1$ only | variable | — |
| Hodgkin–Huxley | 4-d ODE | 4 | $V$ only | variable | — |
| pharmacokinetics (Bateman) | ODE (closed form) | 3 | concentration | variable | log-normal MCMC |
| Fisher–KPP | PDE (64-point grid) | 2 | 3 sensors | variable | PDE-likelihood MCMC |

**Table 1:** The benchmark suite. "obs." lists the observed coordinate when the state is only partially visible; "design" distinguishes the fixed-grid gallery from variable-design systems, where every observation set carries its own set of (time, sensor) observation points. Dashes in the last column indicate systems where no tractable reference exists — these are evaluated by SBC and controls only, which is precisely the regime that motivates carrying exact-reference systems in the same suite.

state), a Poisson-mixture transition density for the Merton jump-diffusion, log-normal residuals around the Bateman curve for pharmacokinetics, and a Gaussian observation model around the deterministic PDE solve for Fisher–KPP — in the last case one likelihood evaluation is one PDE solve.

- *Chain validation.* A reference is used only after validation. Every MCMC reference in this report is run at least twice from well-separated starting points, and the between-chain discrepancy of posterior means (in posterior-sd units) is reported with the result it certifies. Where a closed form and a chain both exist, the chains reproduce it to a worst mean discrepancy of 0.099 sd and width ratios 0.91–1.03 over 24 comparisons.

Twice during development the screen and the record contradicted each other; both times the contradiction exposed an implementation error (a padding mask leaking into set-level statistics; the positional contract of Sec. 4.2). We consider the redundancy a feature of the methodology rather than an inefficiency.

### 6.3 Encoder probing

To separate encoder failures from flow failures, we fit ridge regressions from the pooled encoder output to known sufficient statistics of the problem. High probe $R^2$ with poor posterior accuracy localizes the defect downstream of extraction. In the Merton analysis of Sec. 9, the failing configuration extracted robust volatility statistics *better* than the healthy one ($R^2 = 0.967$ on truncated realized variance), redirecting the repair from the encoder's capacity to the stability of its input representation.

## 7 Benchmark suite

The suite contains fifteen systems (Table 1), defined below with their equations, the meaning and prior range of every parameter, and the observation model. Variable-design training — observation subsets resampled at every optimizer step — is the package default and the recommended mode: beyond enabling inference at any design, the resampling acts as data augmentation over designs. Nine systems are nevertheless benchmarked on conventional fixed grids, for two practical reasons: their closed-form or cheap references live on those grids, and the external tools of Sec. 8.2 consume fixed-length inputs.

All SDEs are integrated by Euler–Maruyama on the stated fine grid; ODEs by RK4 (Hodgkin–Huxley by a semi-implicit scheme for the gating variables); the PDE by finite differences. Systems marked *random designs* are trained and evaluated in the design-amortized mode of Sec. 2: the observation times (and, for the PDE, sensor indices) of every observation set are drawn at random, and one trained network serves all of them; the remaining systems use one conventional fixed grid. The *classical estimator* listed for each system is the standard non-Bayesian method a practitioner would apply; all of them return point estimates, not distributions, and they serve as an accuracy floor for the posterior mean. Three recur and are defined here once. *Kramers–Moyal regression* estimates the drift and squared diffusion from the conditional first and second moments of the observed increments ($f(x) \approx \mathbb{E}[\Delta X \mid X = x]/\Delta t$, $\sigma^2(x) \approx \mathbb{E}[\Delta X^2 \mid X = x]/\Delta t$) and fits the parametric form to these estimates by least squares. *Multi-start nonlinear least squares* fits the deterministic solve of the system to the observed trajectory by minimizing squared error over parameters, restarted from many random initial points to escape local minima. *Exact maximum likelihood* is available where the transition density is known in closed form. Fixed-design observations are token sets over the union of the stated channels; "fast/slow" channels observe every step over a short window and every $j$-th step over the horizon respectively. Priors are uniform on the stated boxes. Classical estimators (`sota()`) are listed per system.

**Linear-Gaussian.** $y = Am + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 0.5^2 I_6)$, with a fixed correlated design matrix $A \in \mathbb{R}^{6\times4}$; $m \in [-3, 3]^4$. Exact Gaussian posterior; the classical estimator is the posterior mean itself. This system checks the flow engine in isolation from the encoder.

**Ornstein–Uhlenbeck.** $dX = -\theta X\, dt + \sigma\, dW$, stationary start $X_0 \sim \mathcal{N}(0, \sigma^2/2\theta)$; $dt = 0.02$, 500 steps; channels: fast ($1\times50$), slow ($20\times24$). Prior $\theta \in [0.3, 3]$, $\sigma \in [0.2, 1.5]$. Classical: exact MLE from per-gap Gaussian transitions.

**Geometric Brownian motion.** $dS = \mu S\, dt + \sigma S\, dW$, $S_0 = 1$; $dt = 0.01$, 500 steps (five "years"); channels: fast ($1 \times 60$), slow ($10 \times 40$). Prior $\mu \in [-0.2, 0.4]$, $\sigma \in [0.1, 0.6]$. Classical: exact MLE on log-returns. Reference: conjugate posterior (Appendix B).

**Cox–Ingersoll–Ross.** $dX = a(b - X)\, dt + \sigma\sqrt{X}\, dW$; $dt = 0.02$, 500 steps; channels as OU. Prior $a \in [0.3, 3]$, $b \in [0.3, 1.5]$, $\sigma \in [0.1, 0.5]$. Classical: Euler pseudo-likelihood MLE (Cox et al., 1985).

**Double well.** $dX = (\theta_1 X - \theta_2 X^3)\, dt + \sigma\, dW$; $dt = 0.01$, 1000 steps, noise strong enough for barrier crossings; channels: fast ($1 \times 80$), slow ($25 \times 36$). Prior $\theta_{1,2} \in [0.5, 3]$, $\sigma \in [0.4, 1.2]$. Classical: Kramers–Moyal regression (defined above).

**Stochastic Lotka–Volterra.** Prey/predator pair with multiplicative noise, $dx_1 = (\alpha x_1 - \beta x_1 x_2)\, dt + 0.1 x_1\, dW_1$, $dx_2 = (\delta x_1 x_2 - \gamma x_2)\, dt + 0.1 x_2\, dW_2$; both components observed; $dt = 0.01$, 600 steps; channels: fast ($1 \times 50$), slow ($15 \times 40$). Prior $\alpha, \gamma \in [0.8, 1.5]$, $\beta, \delta \in [0.4, 1.2]$. Classical: multi-start nonlinear least squares (defined above).

**FitzHugh–Nagumo.** $\dot{v} = v - v^3/3 - w + I + \text{noise}$, $\dot{w} = \epsilon(v + a - bw)$; only $v$ observed at 25 evenly spaced times over horizon 40 ($dt = 0.05$), observation noise 0.05. Prior $a, b \in [0.5, 0.9]$, $\epsilon \in [0.05, 0.3]$, $I \in [0, 0.5]$. Classical: multi-start nonlinear least squares (defined above).

**SEIR with mortality.** Five-compartment stochastic epidemic (two-phase contact rate $\beta_1 \to \beta_2$, incubation $\alpha$, recovery $\gamma_r$, mortality $\gamma_d$); infected and deceased compartments observed at 20 days over a 60-day horizon ($dt = 0.1$), observation noise 0.004. Prior $\beta_1 \in [0.2, 0.6]$, $\beta_2 \in [0.05, 0.3]$, $\alpha \in [0.2, 0.5]$, $\gamma_r \in [0.05, 0.2]$, $\gamma_d \in [0.005, 0.05]$. Classical: nonlinear least squares on $(I, D)$.

**Polynomial-drift SDE.** $dX = (c_0 + c_1 X + c_2 X^2 + c_3 X^3)\, dt + \sigma\, dW$ — drift identification on a dense polynomial basis, the setting of SINDy (Brunton et al., 2016) without its sparsity prior (a sparse-coefficient variant is discussed in Sec. 10); $dt = 0.01$, 1000 steps; channels: fast ($1 \times 80$), slow ($20 \times 45$). Prior $c_0, c_2 \in [-0.5, 0.5]$, $c_1 \in [-2, 0]$, $c_3 \in [-1, -0.1]$, $\sigma \in [0.3, 0.8]$. Classical: least squares of the Kramers–Moyal drift estimate on the polynomial basis — the dense-basis version of SINDy regression.

**Heston (random designs).** $dS = \mu S\, dt + \sqrt{v}\, S\, dW_1$, $dv = \kappa(\bar{v} - v)\, dt + \xi\sqrt{v}\, dW_2$, $\mathrm{corr}(dW_1, dW_2) = \rho$ (Heston, 1993); price observed, volatility latent; $dt = 0.01$, 500 steps, designs of $K \in [2, 128]$ random times. Prior $\mu \in [-0.1, 0.3]$, $\kappa \in [0.5, 5]$, $\bar{v} \in [0.02, 0.3]$, $\xi \in [0.1, 1]$, $\rho \in [-0.9, 0]$.

**Merton jump-diffusion (random designs).** $dS/S = \mu\,dt + \sigma\,dW + d\big(\sum_i (e^{J_i} - 1)\big)$ with Poisson jump rate $\lambda$ and $J_i \sim \mathcal{N}(\mu_J, s_J^2)$ (Merton, 1976); $dt = 0.01$, 500 steps, $K \in [2, 128]$. Prior $\mu \in [-0.1, 0.3]$, $\sigma \in [0.1, 0.5]$, $\lambda \in [0.2, 3]$, $\mu_J \in [-0.3, 0.3]$, $s_J \in [0.05, 0.4]$. Reference: Poisson-mixture transition density (Appendix B).

**Hénon–Heiles (random designs).** Two-dimensional Hamiltonian oscillator with the coupling potential used in Lubich et al. (2015), frequencies $(\omega_1, \omega_2)$ and coupling $\lambda$; $q_1$ observed at random times with noise; $dt = 0.05$, 800 steps, $K \in [4, 64]$. Prior $\omega_{1,2} \in [0.7, 1.3]$, $\lambda \in [0.02, 0.22]$.

**Hodgkin–Huxley (random designs).** Standard four-dimensional membrane model (Hodgkin and Huxley, 1952); voltage observed at random times; parameters $(g_{\mathrm{Na}}, g_K, g_L, I)$ with prior $[60, 180] \times [10, 50] \times [0.05, 0.5] \times [4, 12]$; 3000 integration steps, $K \in [4, 96]$.

**Pharmacokinetics (random designs).** The standard one-compartment oral-dosing model of pharmacokinetics (e.g. Gabrielsson and Weiner, 2000): absorption at rate $k_a$, elimination at rate $k_e$, distribution volume $V$, giving the Bateman concentration curve, concentration $c(t) = \frac{D_0 k_a}{V(k_a - k_e)}\big(e^{-k_e t} - e^{-k_a t}\big)$ with dose $D_0 = 500$; log-normal assay noise (10%); horizon 24 h, $K \in [3, 64]$ irregular draws. Prior $k_a \in [0.5, 4]$, $k_e \in [0.05, 0.4]$, $V \in [20, 100]$. Reference: log-normal likelihood MCMC.

**Fisher–KPP (random designs).** $\partial_t \theta = D\,\partial_{xx}\theta + r\,\theta(1 - \theta)$ on $[0, 1]$ with reflecting boundaries, 64-point finite-difference grid, Gaussian initial bump; three fixed sensors at $x = 16/63, 32/63, 48/63$ observed at random (time, sensor) pairs with noise 0.02; $dt = 0.01$, 200 steps, $K \in [4, 96]$. Prior $D \in [0.001, 0.01]$, $r \in [2, 10]$. Reference: Gaussian observation model around the deterministic solve; one likelihood evaluation is one PDE solve.

**Controls.** Two model-free controls accompany every result. The *prior-mean predictor* ignores the data entirely (exactly 25% mean absolute error under a uniform prior). The *ridge control* is a point predictor of the parameters: a degree-2 ridge regression on generic summary statistics of the token set — per token feature, the mean, standard deviation, mean square, mean absolute value, and the 0.1/0.5/0.9 quantiles, computed over however many observations are present. It recovers no distribution and is not meant to; its role is to detect a specific silent failure that the calibration screen cannot see: a posterior that ignores the data and returns the prior passes SBC perfectly, and only a point-accuracy control unmasks it.

# 8 Experimental results

Two conventions hold throughout. Where a reference posterior exists, accuracy is reported as the signed error of the posterior mean in units of the reference posterior standard deviation (*bias*) and as the ratio of standard deviations (*width*; 1 is exact, above one conservative, below one overconfident). Simulation-based calibration is reported as the number of parameters passing the uniformity screen at $p > 0.05$ over 500 simulated observation sets (*SBC pass*); with 32 such tests on the fixed-grid systems, one or two rejections are expected by chance even from an exact posterior.

## 8.1 Summary

Table 2 collects the outcome for every system: one default configuration, trained per system with no per-system adjustment, passes the calibration screen on 46 of 48 parameters across the suite and matches the exact or validated reference posteriors, where they exist, to within a few percent in width at the largest budgets. The two exceptions are quantified rather than hidden: a $\approx 0.15$ sd location shift on one Cox–Ingersoll–Ross parameter (interval coverage remains 48%/88% against nominal 50%/90%; the identity of the marginal case moves between retrainings, consistent with the resolution limit of SBC at this sample size), and the Merton posterior width, which converges to the reference as the training budget grows (Sec. 8.4). Per-parameter $p$-values, rank distributions, and coverage curves are in Appendix A.

| system | reference | $d$ | SBC pass | worst-param bias / width vs ref | train |
|---|---|---|---|---|---|
| linear-Gaussian | exact posterior | 4 | 4/4 | 0.00 / 1.00 | 11 min |
| Ornstein–Uhlenbeck | exact-likelihood MCMC | 2 | 2/2 | +0.01 / 1.05 | 11 min |
| geometric Brownian motion | conjugate closed form | 2 | 2/2 | −0.00 / 1.05 | 11 min |
| Cox–Ingersoll–Ross | — | 3 | 2/3 | — | 11 min |
| double well | — | 3 | 3/3 | — | 11 min |
| stochastic Lotka–Volterra | — | 4 | 4/4 | — | 11 min |
| FitzHugh–Nagumo | — | 4 | 4/4 | — | 11 min |
| SEIR | — | 5 | 5/5 | — | 11 min |
| polynomial-drift SDE | — | 5 | 5/5 | — | 11 min |
| Heston | — | 5 | 5/5 | — | 12 min |
| Merton jump-diffusion | Poisson-mixture MCMC | 5 | 5/5 | width 1.25 → 1.06 with budget | 8–66 min |
| Hénon–Heiles | — | 3 | 3/3 | — | 8 min |
| Hodgkin–Huxley | — | 4 | 4/4 | — | 9 min |
| pharmacokinetics | log-normal MCMC | 3 | 3/3 | −0.06 / 1.33 at $K = 6$ | 7 min |
| Fisher–KPP (PDE) | PDE-likelihood MCMC | 2 | 2/2 | −0.05 / 1.02 | 6 min |

**Table 2:** All fifteen systems under one default configuration. "SBC pass" $x/y$ means $x$ of the system's $y$ parameters have rank distributions consistent with uniformity ($p > 0.05$; for random-design systems, over mixed designs). The accuracy column quotes the least well-determined parameter against the reference where one exists; dashes mean no tractable reference (calibration screen and controls only). Training time on one H200 GPU; the Merton range spans the budget ladder of Sec. 8.4.

## 8.2 Comparison with existing tools

**Protocol.** We compare against the directly comparable amortized estimators available in maintained packages: NPE with its default masked autoregressive flow (Papamakarios et al., 2017) and FMPE (Wildberger et al., 2023) from `sbi` 0.27 (Boelts et al., 2024); the coupling-flow and flow-matching networks of BayesFlow 2.0 (Radev et al., 2020); and, because it is the closest architecture to ours in the literature, a port of the Simformer (Gloeckler et al., 2024) built from its authors' minimal example (transformer–diffusion over the joint, random condition masks, their architecture and SDE settings, full 75,000-step training). All arms receive the same simulation budget (20,000 draws from the same prior and simulator), run with their package defaults and their own stopping rules, and are compared on exactly three axes: training cost (wall-clock time to fit at this budget), accuracy (posterior-mean bias and width ratio per parameter against the reference posterior, as in Sec. 6.2), and inference cost (wall-clock time per observation set for 2,000 posterior samples). Non-amortized inference sets the scale on both ends: on GBM, whose likelihood is a three-line formula, adaptive Metropolis costs 42 ms per observation set and nothing amortized is worth its training bill for a single instance; on Fisher–KPP, one likelihood evaluation is one PDE solve, so the same reference sampler spends ≈12,000 solves — minutes of computation — on *each* observation set; and for Heston or Hodgkin–Huxley no likelihood formula exists at any price. Because these estimators consume fixed-length vectors, the comparison uses fixed observation designs. Three systems are used, chosen to separate the possible explanations of any performance gap: a multiplicative process where input conditioning is hostile (GBM, exact conjugate reference), an additive process where it is comparatively benign (OU, exact-likelihood MCMC reference, two validated chains per observation set), and a PDE where every likelihood evaluation requires a forward solve (Fisher–KPP), so that the cost relationship between reference and amortized inference is inverted. Where informative, external arms are additionally given hand-derived inputs — log-prices, or per-gap log-returns, the exact coordinates of the GBM likelihood factorization — to separate the effect of the input representation from the generative engine.

**Multiplicative process (GBM).** Table 3 compares all methods at their defaults on raw price data; Table 4 repeats the external methods with hand-derived input coordinates. On raw data every external configuration inflates the volatility posterior — widths 1.85 to 3.16 — irrespective of the generative engine; `sbi` itself emits a warning about precision loss during z-scoring on this data, which is a fair description of the mechanism (Sec. 9). Hand-derived coordinates do not produce convergent behavior (Table 4): with identical log-return inputs, NPE remains 44% over-dispersed, FMPE is nearly exact (1.09), both BayesFlow arms become overconfident (widths 0.42–0.85), and the Simformer port acquires a −0.48 sd bias. Each of these defects is invisible without an exact reference, and each lies within or near the blind region of SBC at practical sizes (Sec. 6). At fixed input coordinates the flow-matching engine outperforms the autoregressive flow (1.09 vs 1.44), consistent with the engine choice of Sec. 3, but no external arm matches the learnable representation operating on raw data (1.05). The verdicts are stable across devices

| method | train | inference | $\mu$: bias / width | $\sigma$: bias / width |
|---|---|---|---|---|
| `amortix` CFM | 488 s | 426 ms | +0.00 / 1.00 | −0.00 / 1.05 |
| `sbi` NPE | 29 s | 13 ms | −0.02 / 1.06 | $+0.45 \pm 0.19$ / 2.64 |
| `sbi` FMPE | 27 s | 370 ms | −0.03 / 1.03 | $+0.20 \pm 0.16$ / 2.46 |
| BayesFlow 2 coupling | 384 s | 4 ms | −0.04 / 0.95 | $-0.13 \pm 0.11$ / 1.85 |
| BayesFlow 2 flow matching | 220 s | 2220 ms | +0.06 / 1.00 | $-0.36 \pm 0.11$ / 1.99 |
| Simformer port* | 591 s | 1057 ms | −0.04 / 1.08 | $-0.30 \pm 0.20$ / 3.16 |
| exact-likelihood MCMC | — | 42 ms | reference (chain check: −0.04 / +0.01) | |

**Table 3:** GBM, all methods on raw price data at package defaults: training cost, accuracy against the exact posterior (bias in reference-sd units; width ratio, 1 exact), and inference cost per observation set for 2,000 samples. Fixed 95-point design, 20,000-simulation budget, 200 test observation sets; timings on one laptop CPU. *Simformer is research code ported from its authors' minimal example and run on an H200 GPU: its timing entries are not device-comparable with the rest, its accuracy entries are.

| method | input given | $\mu$: bias / width | $\sigma$: bias / width |
|---|---|---|---|
| `sbi` NPE | log-prices | −0.04 / 1.01 | $+0.32 \pm 0.17$ / 2.29 |
| `sbi` NPE | log-returns | +0.07 / 1.07 | $+0.13 \pm 0.10$ / 1.44 |
| `sbi` FMPE | log-returns | +0.01 / 1.03 | $-0.02 \pm 0.07$ / 1.09 |
| BayesFlow 2 coupling | log-returns | −0.01 / 0.42 | $-0.15 \pm 0.05$ / 0.54 |
| BayesFlow 2 flow matching | log-returns | −0.04 / 0.46 | $-0.30 \pm 0.06$ / 0.85 |
| Simformer port | log-returns | −0.08 / 1.10 | $-0.48 \pm 0.02$ / 1.36 |

**Table 4:** Input sensitivity of the external methods on GBM: the same tools as Table 3 given hand-derived coordinates — log-prices, or the per-gap log-returns that constitute the exact coordinates of the likelihood factorization. Widths below one are overconfident. `amortix` appears only in Table 3: it receives raw data by construction.

and seeds: re-running the sbi and BayesFlow arms on an H200 GPU with fresh seeds reproduces every failure direction (NPE 2.48, FMPE 2.30, BayesFlow coupling 1.80 on raw input; BayesFlow overconfident at 0.40–0.72 on log-returns).

**Additive control (Ornstein–Uhlenbeck).** The GBM results admit two readings: either the external tools are generally inaccurate, or raw multiplicative data is what breaks them. OU decides between the readings, because it is an additive process with no floating scale. Under the identical protocol (Table 5), the external tools improve exactly as the conditioning explanation predicts. Their volatility bias disappears ($+0.19 \pm 0.19$ sd for NPE, $+0.08 \pm 0.12$ for FMPE, both consistent with zero), and their width inflation drops by roughly half — $2.6 \to 1.9$ for NPE, $2.5 \to 1.6$ for FMPE — but does not vanish: turning 74 raw values into quadratic-variation information is still a representation task, and a z-scored vector input leaves it entirely to the network. `amortix` reads 1.03 and 1.05 on the same observation sets.

**Simulator-limited regime (Fisher–KPP).** The third system inverts the cost structure of the first two: every likelihood evaluation requires solving the forward model. A one-dimensional reaction–diffusion equation is inexpensive in absolute terms (16 ms per solve here); what makes the case informative is the *ratio* between reference and amortized inference, because both sides scale linearly with the solve cost — a chain of $n$ likelihood evaluations costs $n$ solves per observation set whatever the forward model, while the amortized cost per observation set does not involve the simulator at all. The measured ratio below transfers to genuinely expensive simulators, where the same reference sampler would take days per observation set. A single fixed design of $K = 40$ random (time, sensor) points is shared by all observation sets and methods; the reference is the PDE-likelihood sampler of Sec. 6.2 with two validated chains per observation set (271 s per observation set for the pair; two-chain validation over 32 observation sets: worst mean discrepancy 0.22 sd, mean 0.07 sd). Table 6 gives the results. `amortix` matches the reference (1.02/1.04 width, biases within 0.13 sd) at 31 ms per observation set; its entire training cost (379 s) equals the reference cost of fewer than three observation sets, so amortization pays for itself almost immediately in any repeated-inference setting. NPE at defaults shows moderate width inflation (1.26–1.28); FMPE at defaults fails outright on the diffusivity ($+7.3 \pm 2.7$ sd bias at $12.5\times$ width) while remaining reasonable on the reaction rate — a further instance of default configurations behaving discontinuously across parameters, and detectable here only because the expensive reference was computed.

| method (raw values as input) | train | inference | $\theta$: bias / width | $\sigma$: bias / width |
|---|---|---|---|---|
| `amortix` CFM | 183 s | 32 ms | −0.01 / 1.03 | +0.01 / 1.05 |
| `sbi` NPE | 128 s | 10 ms | +0.07 / 1.15 | +0.19 ± 0.19 / 1.88 |
| `sbi` FMPE | 87 s | 147 ms | +0.11 / 1.01 | +0.08 ± 0.12 / 1.60 |
| exact-likelihood MCMC | — | 124 ms | reference (two chains per observation set) | |

**Table 5:** Ornstein–Uhlenbeck, fixed 74-point design, 20,000-simulation budget, 100 test observation sets, one H200 GPU (MCMC on its CPU; time given for the two validation chains). The additive control: external biases vanish and widths improve but remain inflated; compare Table 3.

| method (40 observed values as input) | train | inference | $D$: bias / width | $r$: bias / width |
|---|---|---|---|---|
| `amortix` CFM | 379 s | 31 ms | −0.05 / 1.02 | +0.13 / 1.04 |
| `sbi` NPE | 218 s | 7 ms | +0.19 ± 0.12 / 1.28 | −0.06 / 1.26 |
| `sbi` FMPE | 115 s | 405 ms | +7.3 ± 2.7 / 12.5 | −0.25 ± 0.10 / 1.19 |
| PDE-likelihood MCMC | — | 271 s | reference (two chains per observation set) | |

**Table 6:** Fisher–KPP, fixed $K = 40$ spatio-temporal design, 20,000-simulation budget, 32 test observation sets, one H200 GPU (MCMC on its CPU pool). One likelihood evaluation is one PDE solve, so the reference costs 271 s per observation set; the training cost of the amortized model equals the reference cost of fewer than three observation sets.

**Scope of the comparison.** Every external arm ran at its package defaults under our simulation budget; specialists could improve each with problem-specific embedding networks or summary statistics. The comparison is designed to measure exactly that dependence: what accuracy is delivered when the input representation is not hand-engineered per problem. The Simformer arm in particular is a port of a minimal example rather than a tuned configuration, and its numbers should be read as representative of that recipe, not of the method's ceiling. Conversely, the variable-design capability of Sec. 8.5 is not exercised here at all, since the external estimators consume fixed-length inputs by default.

## 8.3 Posterior approximation

Scalar summaries answer whether the posterior is calibrated on average; Fig. 2 shows the posteriors themselves. For every system with an exact or validated reference, the joint posterior of the trained model is overlaid on the reference for one representative test observation set — three systems with simple posteriors that serve as checks (linear-Gaussian, GBM, OU), and three whose posterior geometry is nontrivial: the Merton jump-diffusion, where volatility and jump rate compete to explain the same variance; pharmacokinetics at six irregular blood draws, the data-poor clinical regime; and Fisher–KPP, whose posterior concentrates on the curved ridge $Dr \approx$ const fixed by the observable front speed. The learned posteriors reproduce location, scale, orientation, and — where present — the curvature of the ridge.

The joint agreement can be summarized in one number per system. Following the construction used as the FID score in the image literature, we fit Gaussians to the two sample sets in reference-normalized coordinates (each parameter divided by the reference posterior standard deviation) and report the squared Fréchet distance

$$d_F^2 \;=\; \|\mu_a - \mu_r\|^2 + \operatorname{tr}\!\Big(\Sigma_a + \Sigma_r - 2\,(\Sigma_r^{1/2}\Sigma_a\Sigma_r^{1/2})^{1/2}\Big), \tag{10}$$

which is zero for a perfect match, grows as squared bias in units of reference sd, and penalizes covariance (orientation) errors that per-parameter width ratios miss. Averaged over the test observation sets of Fig. 2: linear-Gaussian 0.03, OU 0.04, GBM 0.08, Fisher–KPP 0.16, Merton 0.54 (dominated by one test set with $\sigma$ width 1.24; the other two sit at 0.1–0.2), pharmacokinetics 2.1. For scale, a 0.1 sd bias in each of $d$ dimensions gives $d_F^2 = 0.01\,d$; the pharmacokinetics value is the one clear misfit, and it is a budget effect — the next subsection returns to it. Together with Table 2, this figure is the central evidence that one small default model recovers full posteriors across the suite.

## 8.4 Convergence and computational cost

Accuracy against the exact references is a smooth, monotone function of the training budget, and we report it systematically rather than at a single operating point. Figure 3 shows posterior width and bias against

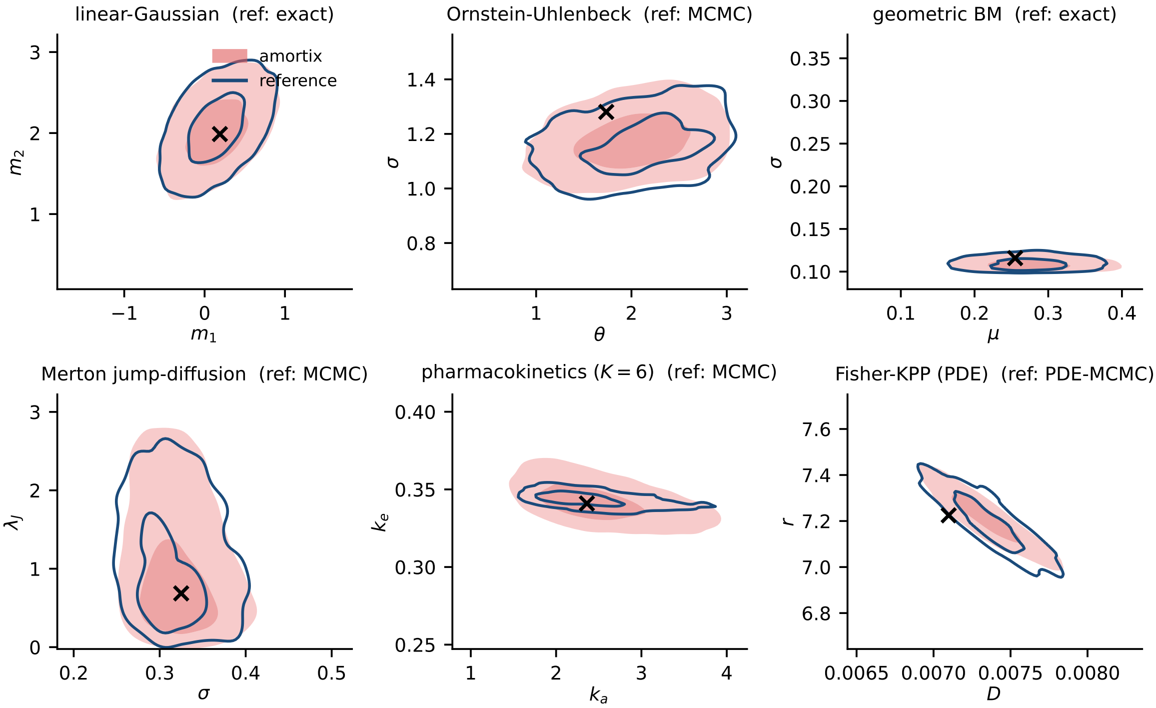


**Figure 2:** Joint posteriors of `amortix` (filled contours: 39% and 86% highest-density regions) against exact or validated-MCMC reference posteriors (open contours, same levels) on one test observation set per system; crosses mark the true parameters. Top row: systems with closed-form or cheap exact references. Bottom row: systems where the reference itself required exact-likelihood MCMC with two-chain validation — including the ridge-shaped Fisher–KPP posterior, where the model reproduces the curvature, not just the moments.

the reference for three systems (GBM, OU, Fisher–KPP with the fixed design of Sec. 8.2) over a grid of budgets from 5,000 to 40,000 simulations with optimizer steps scaled proportionally. Posterior widths converge to one from above on all three systems (Fisher–KPP: $1.45 \to 1.01$). Posterior-mean biases behave differently by identifiability: on GBM and OU they stay within $\pm 0.1$ sd throughout, while on Fisher–KPP the bias lies along the weakly identified ridge direction and does not decay monotonically — independently trained models land at different points within $\pm 0.5$ sd along the ridge, consistent with the flat direction carrying the weakest training signal (Sec. 3). Under the variable-design training procedure, which averages every optimizer step over fresh observation designs, the same ridge bias does decay with budget. Beyond it, the two residual defects of the variable-design suite — inflated width on the hardest Merton case and along-ridge bias on Fisher–KPP under random designs — respond to the same budget lever (Fig. 4). The Merton width excess follows $\propto 1/\text{budget}$ ($0.246 \to 0.120 \to 0.059$ at $\times 1, \times 3, \times 6$); the KPP ridge bias follows $\approx \text{budget}^{-1/2}$ ($+0.41 \to +0.23 \to +0.14$ sd along the ridge, with the sharply identified front-speed component decaying in parallel). Two negative controls exclude the obvious alternatives: doubling network width and depth at fixed budget changes the Merton case by less than 0.02 in width ratio (1.103 vs 1.120 at $\times 3$), and refining the sampling ODE solver from 20 midpoint steps to 60 RK4 steps changes nothing measurable. The weakly identified posterior direction carries the weakest regression signal in (3) and converges last; we find no evidence of a floor within the measured range.

Costs, for orientation. Training takes 6–12 minutes per system at the base budget on one H200 GPU (Table 2) and about an hour at the $\times 6$ budget behind the sharpest Merton numbers; on a laptop CPU, base-budget training takes 8–45 minutes and produces bit-identical posteriors, because all stochastic generation is kept on a CPU random number generator by design. Inference draws 2,000 posterior samples in 426 ms per observation set on the laptop CPU and 72 ms batched (135 ms single) on the GPU. The small external models of Sec. 8.2 train and sample faster at their defaults; the tiny-model regime is kernel-launch-bound on GPUs (`sbi`'s default flow trains $6\times$ slower there than on the laptop CPU).

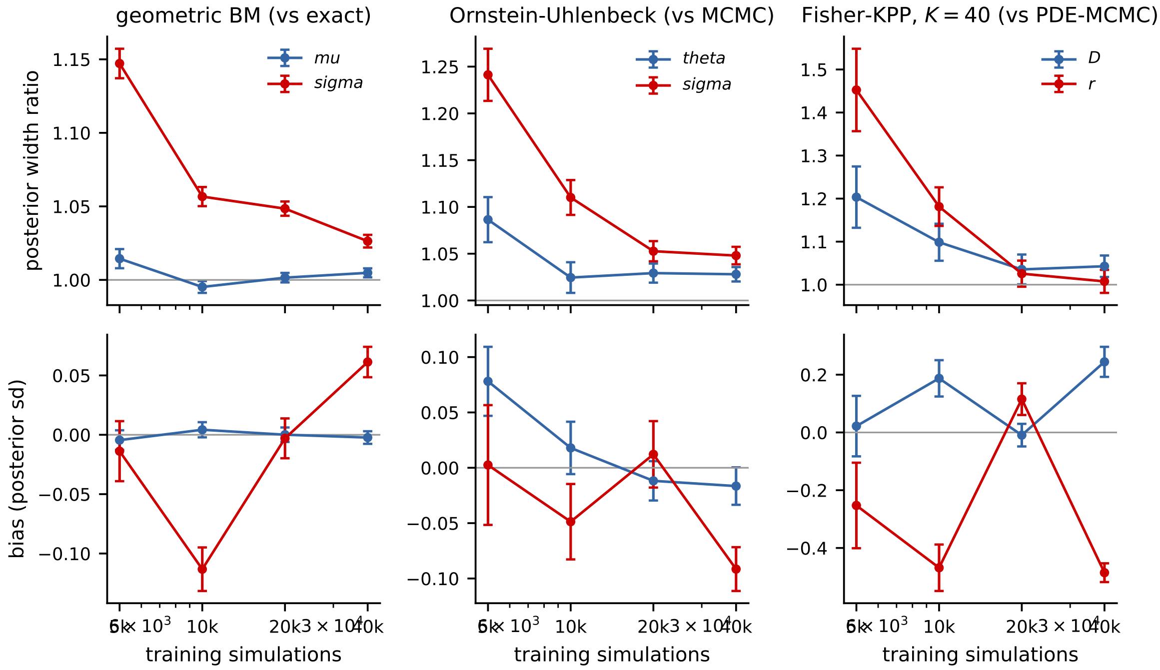


**Figure 3:** Convergence to the reference posterior with training budget on three systems (columns): posterior width ratio (top) and posterior-mean bias (bottom) per parameter, over budgets of 5,000–40,000 simulations with optimizer steps scaled proportionally. References: exact conjugate posterior (GBM), exact-likelihood MCMC (OU), PDE-likelihood MCMC (Fisher–KPP, fixed $K = 40$ design).

### 8.5 Variable designs and ablations

**Variable-design zoo.** With the class rule and training procedure fixed, six further systems were trained without per-system adjustments. Heston (five parameters, latent volatility): all 25 SBC cases pass, and pair tokens give no advantage over point tokens, consistent with the class boundary (Sec. 4.3). Merton jump-diffusion: 25 of 25 cases after the heavy-tail treatment of Sec. 9, including dense-design cases that failed at $p < 10^{-3}$ before it; the residual width sits on the budget curve below. Hénon–Heiles (parameters of the potential of Lubich et al. 2015, observing one coordinate at random times): calibrated at every design size down to $K = 6$, where the posterior widens honestly rather than degrading. Hodgkin–Huxley (Hodgkin and Huxley, 1952): calibrated on mixed designs; one dense-design case shows the budget signature. Pharmacokinetics: at $K = 6$ irregular blood draws, bias $-0.01$ to $-0.06$ sd and widths 1.09–1.33 against exact-likelihood MCMC. Fisher–KPP: the posterior concentrates on the ridge $D \cdot r \approx$ const; the residual along-ridge bias decays with budget (next subsection).

Table 7 and Fig. 5 compare encoder variants for design amortization on GBM, trained at $K \sim$ log-uniform$[2, 128]$ random-time designs and evaluated against the exact per-design posterior (96 observation sets per bucket). Three observations organize the table. First, index-based positions combined with in-place subsampling during training (leaving kept tokens at spread positions while inference packs contiguously) produce biases of $+1.1$ to $+3.3$ posterior sd with no individually incorrect component — the positional pathology of Sec. 4.2; repacking or replacing positions with time phases removes the bias entirely. Second, all point-token variants share inflated widths at dense designs, in proportion to how poorly their pooled representation encodes $\log K$ (rightmost column; the probe of Sec. 4.3) — including two attention mechanisms specifically constructed to exploit continuous time (a learned per-head time-kernel logit bias in the spirit of relative-position methods, and edge-message attention carrying gap-modulated increments on all pairs). Third, pair tokens with time-phase attention, plus the training procedure of Sec. 5, bring every bucket to within 7% of the exact width.

# 9 Failure analysis: conditioning of the encoder input

Every accuracy failure encountered during the development of this package — including, we argue, the external failures of Table 3 — reduces to a single cause appearing in three forms: the conditioning of

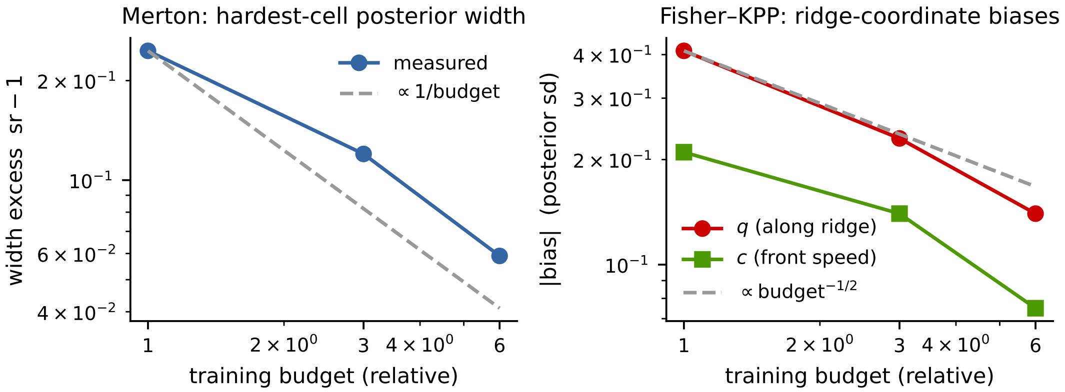


**Figure 4:** The two residual defects of the suite against training budget (simulations and steps scaled together; model size fixed). Left: Merton width excess on the hardest evaluation case, with the 1/budget guide. Right: Fisher–KPP posterior-mean biases in the ridge coordinate system, with the budget$^{-1/2}$ guide.

| | $\sigma$ width ratio at $K$ | | | | |
|---|---|---|---|---|---|
| encoder variant | 5 | 9 | 20 | 100 | $R^2(\log K)$ |
| point tokens, index positions, spread subsampling | 0.94* | 1.13* | 1.42* | 2.08* | — |
| point tokens, index positions, contiguous | 1.05 | 1.17 | 1.13 | 1.73 | 0.957 |
| point tokens, time-kernel attention bias | 1.03 | 1.14 | 1.31 | 2.51 | 0.712 |
| point tokens, edge-message attention | 1.03 | 1.15 | 1.29 | 2.45 | 0.498 |
| point tokens, time phases | 1.01 | 1.10 | 1.14 | 1.95 | — |
| + explicit $\log K$ feature | 1.01 | 1.05 | 1.08 | 1.49 | — |
| pair tokens, time phases | 1.02 | 1.02 | 1.02 | 1.27 | 0.985 |
| pair tokens + full training procedure | 1.06 | 1.06 | 1.05 | 1.07 | — |

**Table 7:** GBM with variable designs: posterior width for the volatility $\sigma$ against the exact per-design posterior (96 observation sets per design-size bucket) and, where measured, the probe $R^2$ for $\log K$ from the pooled encoder output. The table tracks $\sigma$ because it is the parameter whose posterior width is governed by the observation density — the drift $\mu$ is weakly informed at every $K$, and all variants recover its posterior within a few percent, so it does not discriminate between architectures. *This row also carries biases of +1.09 to +3.30 posterior sd (the positional train/test mismatch of Sec. 4.2); all other rows have biases consistent with zero.

the numbers presented to the set encoder. We document the three forms with their measurements and repairs; all three repairs are learnable modules rather than problem-specific transformations.

**Form 1: floating multiplicative scale.** On GBM, raw price tokens force the network to divide squared increments by squared levels internally; with global standardization this manifests as a volatility bias of +0.48 posterior sd (confirmed against the closed-form posterior) that no amount of training removes. The repair is the monotone mixture reparametrization of Sec. 4.3, which learns the log coordinate; the Yeo–Johnson ablation there shows why the parametrization, not the hypothesis class, is the operative choice. External estimators exhibit the same form on the same data (Table 3, raw-price rows).

**Form 2: cardinality blindness.** Normalized attention erases set size, which controls posterior width under design amortization; the probe measurements and the $\log K$/register-token repairs are given in Sec. 4.3, with the arm-by-arm consequences in Table 7.

**Form 3: heavy tails.** On the Merton jump-diffusion, squared increments have sample standard deviation $3.1 \times 10^4$ against a diffusion signal of order $10^{-2}$–1; running feature normalization never converges, and the flow trains against a drifting representation, producing a +1.12 sd volatility bias. Encoder probing (Sec. 6.3) shows the information is extracted — the failing configuration encodes robust volatility statistics better than the healthy one — so the repair targets representation stability: logarithmic compression of increment features (near-identity on clean diffusions since $\log(1 + x) \approx x$ for small $x$) restores all 25 evaluation cases. A precision follow-up against the Poisson-mixture reference then isolated a smaller (+0.27 sd) residual whose mechanism is a jump-classification threshold that floats with the

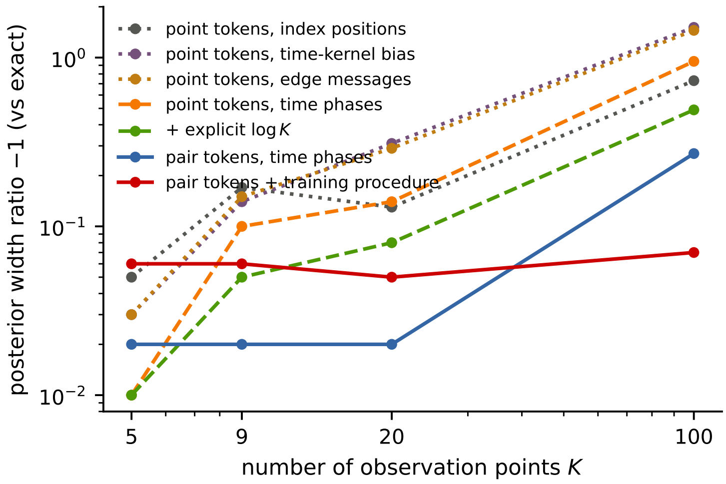


**Figure 5:** The data of Table 7 in graphical form: excess posterior width for the GBM volatility — the width ratio against the exact posterior minus one, so 0.1 means a posterior 10% too wide and lower is better — as a function of the number of observation points $K$ (both axes logarithmic). Dotted curves: point-token variants with no set-size information, which degrade as designs become dense; dashed: partial repairs; solid: increment tokens, and increment tokens combined with the training procedure of Sec. 5, which is flat in $K$.

unknown volatility; a per-observation-set set-conditioning module (pooled statistics modulating feature channels, initialized to the identity) removes it as effectively as a hand-crafted per-observation-set rescaling ($+0.127 \pm 0.055$ vs $+0.094 \pm 0.054$ sd, statistically indistinguishable). This was one of three such paired comparisons in the project (log coordinate, set size, robust scaling); in all three the learnable module matched the hand-crafted repair once initialized at the working configuration. We summarize the pattern as a design principle: the hand-crafted fix identifies the mechanism and the initialization; the learnable module placed there is what ships.

Three implementation conditions proved necessary for the set-conditioning module and are recorded here because each was found through a failure: the modulation must be zero-initialized (training must start from the working baseline); it must act after feature normalization, not before (the reverse ordering re-creates the representation drift it is meant to cure); and pooled set statistics must exclude padding positions (a masking error here made padded and packed representations of the same observation set inequivalent, detected only because SBC and the exact-reference probe disagreed).

# 10 Discussion and limitations

The evidence assembled here supports a specific claim: for dynamical-systems inverse problems, the input representation of an amortized posterior estimator is not a preprocessing detail but the dominant accuracy factor, and it can be made part of the model rather than part of the problem specification. The comparison of Sec. 8.2 shows the cost of the alternative — fixed representations with global standardization — across four generative engines, on a problem where the correct answer is known in closed form. The OU control separates the mechanisms: on additive data the bias of the external estimators disappears and their width inflation drops by roughly a factor of two but does not vanish, so the conditioning pathology accounts for the dominant part of the failure and a smaller representation-learning gap accounts for the rest.

Limitations. (i) The remaining inaccuracies of the suite decay as measured power laws in training compute (Fig. 3); we have not searched for methods that improve the exponents. (ii) The power limits of SBC imply that small residuals can only be certified where an exact or validated reference exists; systems without tractable likelihoods inherit the weaker screen, and carrying reference-equipped systems alongside them in one suite is our current mitigation. (iii) The comparison of Sec. 8.2 uses package defaults; it measures the out-of-the-box regime, not the ceiling of any method. (iv) Bayesian experimental design — the intended consumer of design amortization — is not yet demonstrated; the training procedure of Sec. 5 was built so that the required object $p(m \mid S)$ is already coherent across subsets $S$. (v) No real-data study with posterior predictive checks is included yet. (vi) The polynomial-drift system uses a dense coefficient prior; the scientifically interesting variant is a *sparse* prior — train on coefficient vectors with random supports and test whether the posterior recovers the support — which requires extending the

| system | per-parameter SBC $p$-values |
|---|---|
| linear-Gaussian | $m_1$: 0.06  $m_2$: 0.60  $m_3$: 0.38  $m_4$: 0.56 |
| Ornstein–Uhlenbeck | $\theta$: 0.28  $\sigma$: 0.77 |
| SEIR | $\beta_1$: 0.33  $\beta_2$: 0.22  $\alpha$: 0.61  $\gamma_r$: 0.34  $\gamma_d$: 0.53 |
| geometric Brownian motion | $\mu$: 0.26  $\sigma$: 0.44 |
| Cox–Ingersoll–Ross | $a$: 0.65  $b$: **0.001**  $\sigma$: 0.73 |
| double well | $\theta_1$: 0.78  $\theta_2$: 0.54  $\sigma$: 0.88 |
| stochastic Lotka–Volterra | $\alpha$: 0.07  $\beta$: 0.69  $\delta$: 0.08  $\gamma$: 0.35 |
| FitzHugh–Nagumo | $a$: 0.19  $b$: 0.15  $\epsilon$: 0.90  $I$: 0.63 |
| polynomial-drift SDE | $c_0$: 0.06  $c_1$: 0.96  $c_2$: 0.46  $c_3$: 0.80  $\sigma$: 0.50 |

**Table 8:** Per-parameter SBC uniformity $p$-values for the fixed-design gallery run of Table 2 (500 observation sets $\times$ 200 draws; $p > 0.05$ passes). The single failure is marked in bold; see Sec. 8.1 for its behavior across retrainings.

normalization (2) to priors with atoms at zero, a planned extension of the package. (vii) All priors are uniform on boxes; the probit normalization (2) extends to any factorized prior with tractable CDFs, but correlated priors would require a different normalization.

# A Calibration detail for the fixed-grid systems

Per-parameter SBC $p$-values for the run of Table 2 are listed in Table 8; Fig. 6 shows empirical-vs-nominal coverage of the central credible intervals, and Fig. 7 the underlying rank distributions.

# B Reference posteriors

**GBM conjugate posterior.** With observations at indices $i_0 = 0 < i_1 < \dots$ of the fine grid, log-returns $r_k = \log(S_{i_{k+1}}/S_{i_k})$ over gaps $\tau_k$ are independent $\mathcal{N}\big((\mu - \sigma^2/2)\tau_k, \sigma^2\tau_k\big)$. In the coordinates $(b, \sigma^2) = (\mu - \sigma^2/2, \sigma^2)$ the model is a Gaussian location–scale family with sufficient statistics $T = \sum_k \tau_k$, $R_1 = \sum_k r_k$, and $SS_w = \sum_k r_k^2/\tau_k - R_1^2/T$; draws from the standard conjugate posterior ($\sigma^2 = SS_w/\chi^2_{n-1}$, $b \mid \sigma^2 \sim \mathcal{N}(R_1/T, \sigma^2/T)$) are transformed to $(\mu, \sigma)$ and importance-resampled to the uniform box prior with the appropriate Jacobian weight. The sampler was validated against exact-likelihood MCMC on the same designs: over 24 comparisons, worst posterior-mean discrepancy 0.099 sd, width ratios 0.91–1.03.

**Adaptive Metropolis.** Random-walk Metropolis on the box, proposal $x' = x + s\,C\,\xi$, $\xi \sim \mathcal{N}(0, I)$. During burn-in (default $\max(500, n/2)$ iterations) the scalar $s$ is adapted toward a 0.25 acceptance rate in blocks of 50 with a decaying Robbins–Monro gain, and once sufficient burn-in states are collected the shape $C$ is replaced by the Cholesky factor of the empirical covariance rescaled by $2.38/\sqrt{d}$ (Haario et al., 2001); adaptation is frozen for the retained phase. Out-of-box proposals are rejected, which samples the box-truncated target exactly since the proposal is symmetric. Likelihoods: per-gap Euler–Maruyama Gaussian transitions for OU (matching the generative process, plus the stationary density of $X_0$); the exact GBM log-return density; the Poisson-mixture transition density for Merton (truncated at the numerically negligible jump count); log-normal residuals around the Bateman curve; a Gaussian observation model around the deterministic PDE solve for Fisher–KPP. Every chain used as a reference is paired with a second chain from a well-separated start; between-chain posterior-mean discrepancies are reported alongside the results they certify (Sec. 8.2).

## C Correspondence with package options

| in this report | in the package |
|---|---|
| point tokens | `embed="wpoint"` |
| pair tokens + set conditioning | `embed="wfilm"` |
| warped values/increments (fixed designs) | `embed="wbasis"` / `"wdiff"` |
| attention phases from physical time | `rope="time"` |
| fresh designs each optimizer step | `fit(retokenize=...)` via `make_retokenizer()` |
| variable-design problem wrapper | `amortix.designs.DesignProblem` |
| benchmark systems | `amortix.problems` (`DESIGN_ZOO`) |
| calibration screen / reference machinery | `sbc_design` / `amortix.mcmc` + likelihood factories |
| comparison scripts (Sec. 8.2) | `examples/baseline_{npe,ou,kpp,bayesflow,simformer}.py` |

All defaults named in this report are selected automatically by `FlowPosterior(problem)`; the non-default options exist as reproducible controls for the comparisons above.

## References


J. Boelts, M. Deistler, M. Gloeckler, et al. sbi reloaded: a toolkit for simulation-based inference workflows. *arXiv:2411.17337*, 2024.

S. L. Brunton, J. L. Proctor, J. N. Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *PNAS* 113(15), 2016.

J. C. Cox, J. E. Ingersoll, S. A. Ross. A theory of the term structure of interest rates. *Econometrica* 53(2), 1985.

K. Cranmer, J. Brehmer, G. Louppe. The frontier of simulation-based inference. *PNAS* 117(48), 2020.

T. Darcet, M. Oquab, J. Mairal, P. Bojanowski. Vision transformers need registers. *ICLR*, 2024.

J. Gabrielsson, D. Weiner. *Pharmacokinetic and Pharmacodynamic Data Analysis: Concepts and Applications*. Swedish Pharmaceutical Press, 2000.

M. Gloeckler, M. Deistler, C. Weilbach, F. Wood, J. H. Macke. All-in-one simulation-based inference. *ICML*, 2024.

D. Greenberg, M. Nonnenmacher, J. Macke. Automatic posterior transformation for likelihood-free inference. *ICML*, 2019.

H. Haario, E. Saksman, J. Tamminen. An adaptive Metropolis algorithm. *Bernoulli* 7(2), 2001.

J. Hermans, A. Delaunoy, F. Rozet, A. Wehenkel, V. Begy, G. Louppe. A trust crisis in simulation-based inference? Your posterior approximations can be unfaithful. *TMLR*, 2022.

S. L. Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Rev. Financial Studies* 6(2), 1993.

A. L. Hodgkin, A. F. Huxley. A quantitative description of membrane current and its application to conduction and excitation in nerve. *J. Physiology* 117(4), 1952.

M. Horn, M. Moor, C. Bock, B. Rieck, K. Borgwardt. Set functions for time series. *ICML*, 2020.

Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, M. Le. Flow matching for generative modeling. *ICLR*, 2023.

C. Lubich, I. V. Oseledets, B. Vandereycken. Time integration of tensor trains. *SIAM J. Numer. Anal.* 53(2), 2015.

J.-M. Lueckmann, J. Boelts, D. Greenberg, P. Gonçalves, J. Macke. Benchmarking simulation-based inference. *AISTATS*, 2021.

R. C. Merton. Option pricing when underlying stock returns are discontinuous. *J. Financial Economics* 3(1–2), 1976.

G. Papamakarios, I. Murray. Fast $\epsilon$-free inference of simulation models with Bayesian conditional density estimation. *NeurIPS*, 2016.

G. Papamakarios, T. Pavlakou, I. Murray. Masked autoregressive flow for density estimation. *NeurIPS*, 2017.

E. Perez, F. Strub, H. de Vries, V. Dumoulin, A. Courville. FiLM: Visual reasoning with a general conditioning layer. *AAAI*, 2018.

W. Peebles, S. Xie. Scalable diffusion models with transformers. *ICCV*, 2023.

S. T. Radev, U. K. Mertens, A. Voss, L. Ardizzone, U. Köthe. BayesFlow: Learning complex stochastic models with invertible neural networks. *IEEE Trans. Neural Netw. Learn. Syst.* 33(4), 2020.

S. N. Shukla, B. M. Marlin. Multi-time attention networks for irregularly sampled time series. *ICLR*, 2021.

J. Su, M. Ahmed, Y. Lu, S. Pan, W. Bo, Y. Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing* 568, 2024.

S. Talts, M. Betancourt, D. Simpson, A. Vehtari, A. Gelman. Validating Bayesian inference algorithms with simulation-based calibration. *arXiv:1804.06788*, 2018.

J. Wildberger, M. Dax, S. Buchholz, S. R. Green, J. H. Macke, B. Schölkopf. Flow matching for scalable simulation-based inference. *NeurIPS*, 2023.

I.-K. Yeo, R. A. Johnson. A new family of power transformations to improve normality or symmetry. *Biometrika* 87(4), 2000.

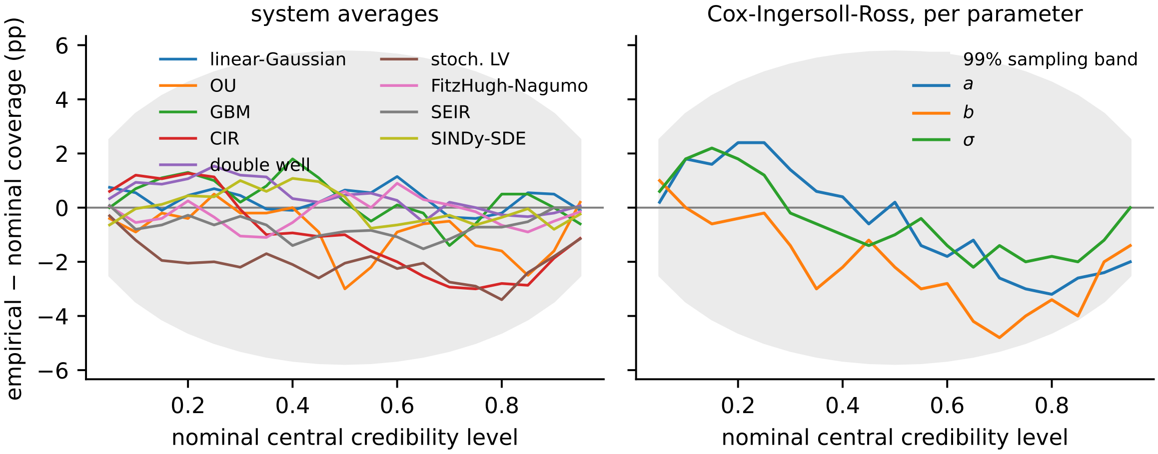


**Figure 6:** Empirical coverage of central credible intervals for the fixed-design gallery, shown as the deviation from the nominal level. For each system, 500 synthetic observation sets are simulated with known parameters and the posterior's central $q\%$ interval is checked against the truth; a calibrated posterior has zero deviation at every level, and the gray band is the 99% pointwise range induced by the finite number of test observation sets. Left: deviation averaged over each system's parameters. Right: individual parameters of Cox–Ingersoll–Ross, the system with the largest deviation in this run; the $b$ curve reflects the $\approx 0.15$ sd location shift discussed in the text. Rank-level detail for all 32 parameters is in Appendix A (Fig. 7).

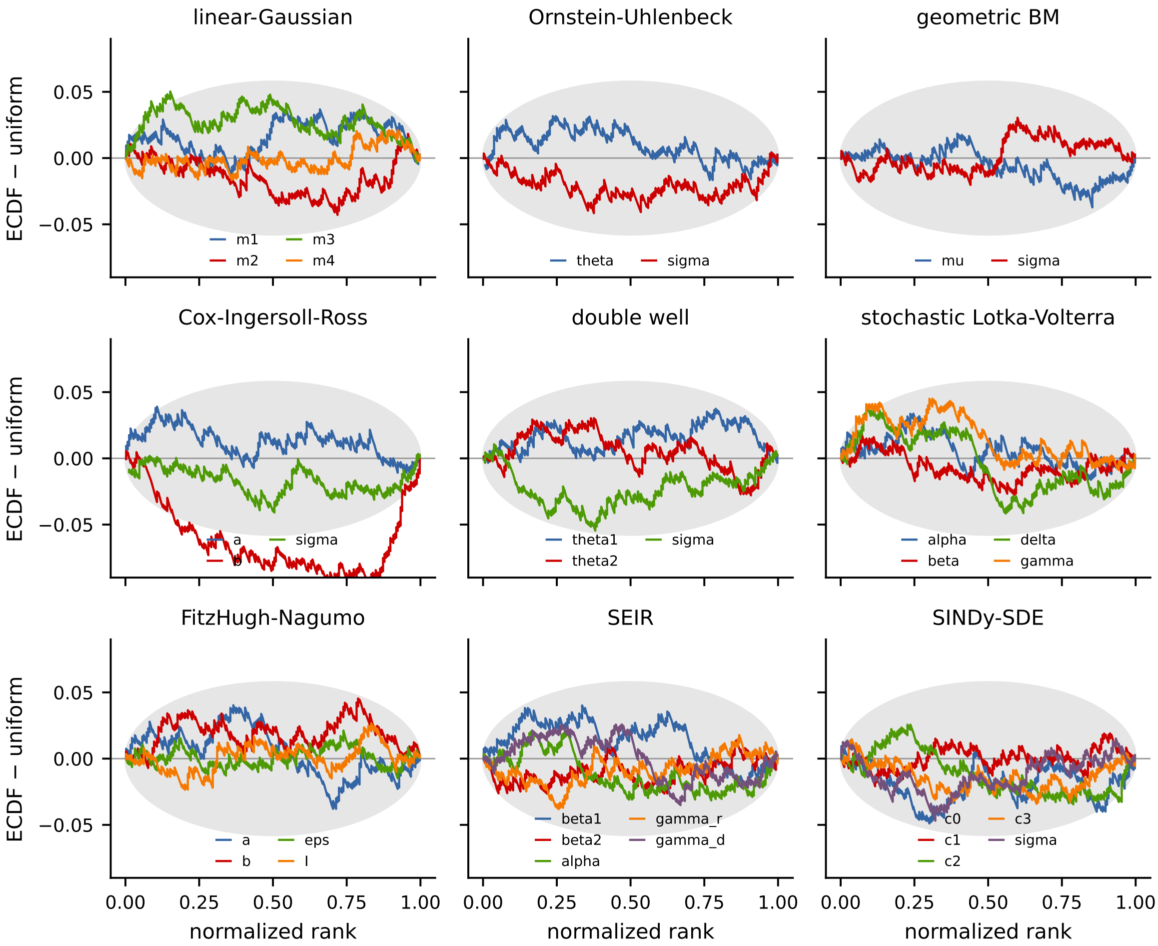


**Figure 7:** Calibration check for the fixed-design gallery. Procedure: for each system, 500 synthetic observation sets are simulated with known parameters; for each observation set the model produces 200 posterior draws, and the rank of the true parameter among the draws is recorded. If the posterior is calibrated, these ranks are uniform. Each curve shows, for one parameter, the deviation of the empirical CDF of normalized ranks from the uniform diagonal; the gray band is the 99% range attributable to sampling noise at 500 observation sets. A curve bulging below zero on the left indicates a posterior shifted low relative to the truth, and vice versa. All curves stay within the band except Cox–Ingersoll–Ross $b$, whose deviation corresponds to the $\approx 0.15$ sd location shift discussed in the text.